

recent television commercial, is becoming the new way to purchase goods and services, transfer funds, and perform other types of wireless transactions.

The migration from simple voice communication to data communication is underway in earnest. According to Cahners In-Stat Group2, the most successful wireless data system is the short message service (SMS) on Global System for Mobile Communications (GSM) networks. Cahners points out that in a single month early this year, users sent 8 billion SMS messages worldwide. Interestingly, for many users in some parts of the world (including most notably, Japan), the wireless device is the most prevalent mode of accessing the Internet, compared to PC Internet access. More than 200 million SMS subscribers already dot the globe, and Cahners projected 742 million worldwide wireless Internet subscribers in 2004 and 607 million SMS subscribers in the same year.

Overview of Threats and Potential Damage

Yet, like each new communication and computing medium before it, wireless voice and data communication presents the opportunity for less desirable applications. The rapid spread of wireless communications presents new opportunities for hackers, disgruntled employees, and others to prove their prowess in spreading viruses and malicious code.

On the surface, the vulnerability of wireless devices to viruses and malicious code threats appears to follow the same patterns of vulnerabilities that the wired world has experienced. Yet, upon closer inspection, the vulnerabilities are more numerous and complex. Such threats can be categorised into three groups:

- Application-based threats
- Content-based threats
- Mixed threats (a power-packed combination of application and content-based threats not yet seen in the real world)

Application-based Threats

In the wireless world, application-based threats are posed by exe-